

Summary of Unit-III

NETWORK LAYER

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Outline

I. Introduction to Network Layer

- *Network Layer Services*
- *Packet Switching*
- *Network Layer Performance*
- *IPv4 Addresses*
- *Forwarding of IP Packets*

II. Network Layer Protocols

- *IPv4*
- *ICMP*

III. Unicast Routing

- *Routing Algorithms*
- *Unicast Routing Protocols*

IV. Multicast Routing

V. Next Generation IP

- *IPv6 Addressing*
- *IPv6 protocol*
- *ICMPv6 protocol*
- *Transition from IPv4 to IPv6*

Summary- *I. Introduction to Network Layer*

- Introduced the network layer by defining the services provided by this layer with packetizing. Described forwarding and routing and compares the two. Also, briefly explained the other services such as flow, error, and congestion control.
- Discussed packet switching, which occurs at the network layer. The datagram approach and the virtual-circuit approach of packet switching are described in detail in this section.
- Discussed network-layer performance. Described different delays that occur in network-layer communication. It also mentions the issue of packet loss. Finally, discussed the issue of congestion control at the network layer.

Summary- *I. Introduction to Network Layer*

- Discussed IPv4 addressing, probably the most important issue in the network layer. Described the address space and briefly discussed classful addressing, which belongs to the past but is useful in understanding classless addressing. Explained classless addressing and several issues related to this topic.
- Discussed DHCP, which can be used to dynamically assign addresses in an organization. Also, discussed NAT, which can be used to relieve the shortage of addresses to some extent.
- Discussed forwarding of network-layer packets and shown how forwarding can be done based on the destination address in a packet using a label.

Summary- *II. Network Layer Protocols*

- Discussed the IPv4 protocol with its datagram format. Explained the purpose of fragmentation in a datagram. Briefly discussed options fields and their purpose in a datagram.
- Discussed ICMPv4, one of the auxiliary protocols used in the network layer to help IPv4. Briefly discussed the purpose of each option and shown how ICMP can be used as a debugging tool. Illustrated how the checksum is calculated for an ICMPv4 message.

Summary- *III. Unicast Routing*

- Introduced the concept of unicast routing and described the general ideas behind it. Least-cost routing and least-cost trees were also discussed.
- Described the common routing algorithms used in Internet. Explained the data transmission in distance-vector routing, link-state routing and path-vector routing.
- Described RIP, a protocol that implements the distance-vector routing algorithm, OSPF, a protocol that implements the link-state routing algorithm and finally, described the BGP, a protocol that implements the path-vector routing algorithm.

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Summary- *IV. Multicast Routing*

- Introduced the concept of multicasting and compares unicasting and multicasting. Explained the differences between multicasting and multiple unicasting. Also, discussed some applications of multicasting.
- Gave some information about the basics of multicasting and elaborated on multicast addresses. Explained the mapping of network layer multicast addresses to data-link layer multicast addresses. Describes how collection of information to be used for forwarding is done differently in unicasting and multicasting. Finally described the process of multicast forwarding and two approaches in multicasting.

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Summary- *VI. Next Generation IP*

- Discussed the addressing mechanism in the new generation of the Internet, its representation and allocation in the address space. Explained auto-configuration and renumbering, which makes it easy for a host to move from one network to another.
- Discussed IPv6 protocol. First the new packet format is described. Then shown how use of extension headers can replace the options.
- Discussed ICMPv6 and how the new protocol replaces several auxiliary protocols in version 4. Also divided the messages in this protocol into four categories and described them.
- Briefly shown how transition can be made from the current version to the new one smoothly. Explained three strategies that need to be followed for this smooth transition.

Self Assessment Questions

1. You are given the following address: 15.16.193.6/21. Determine the subnet mask, address class, subnet address, and broadcast address.

Self Assessment Questions

2. Which one of the following is not a function of network layer?

- a) routing
- b) inter-networking
- c) congestion control
- d) error control

Self Assessment Questions

3. A 4 byte IP address consists of _____

- a) only network address
- b) only host address
- c) network address & host address
- d) network address & MAC address

Self Assessment Questions

4. The network layer protocol for internet is _____

- a) ethernet
- b) internet protocol
- c) hypertext transfer protocol
- d) file transfer protocol

Thank You